PRACTICAL 1

Design webpage that make use of

a. Document Structure Tags

Document structure tags are HTML codes in angle brackets that build the foundational skeleton of a web page. They tell the web browser how to organize the document type, metadata, and visible content.

Core HTML Skeleton Tags

- <!DOCTYPE html>: Tells the browser this is an HTML5 document. It is a declaration, not a true HTML tag.
- <html>: Acts as the root container holding every other element on the page.
- <head>: Holds hidden metadata, links to style sheets, and scripts.
- <title>: Sets the text shown on the browser tab.
- <body>: Holds all visible elements

shown on the screen.

Semantic Content Layout Tags

- <header>: Marks the top banner or introductory menu of a page or section.
- <nav>: Holds major website navigation links.
- <main>: Wraps the unique, core content of the document.
- <section>: Splits content into thematic groups or chapters.
- <article>: Holds independent, self-contained stories or posts.
- <aside>: Holds sidebars and extra related notes.
- <footer>: Marks the bottom area containing copyrights or contact info.

b. Various Text Formatting Tags

HTML text formatting tags are elements used to modify the visual appearance or semantic meaning of text on a webpage.

1. Heading Tags

Heading tags are used to define the titles and subtitles of a webpage, ranging from <h1> (most important) to <h6> (least important).

2. Bold and Importance
   - <b>: Makes text **bold** purely for visual purposes without adding extra importance.
   - : Displays text in **bold** while signaling to browsers and search engines that the content holds high semantic importance.
3. Italics and Emphasis
   - <i>: Italicizes text purely for visual styling (e.g., technical terms or idiomatic phrases).
   - : Italicizes text while adding **semantic emphasis**, modifying the oral meaning for screen readers.
4. Alterations and Corrections
   - <u>: Underlines text visually.

- <del>: Strikingly crosses out text to signify **deleted** or outdated content.
- <ins>: Underlines text to highlight **inserted** or newly added content.

5. Sizing and Positioning

- <small>: Decreases the font size by one level relative to the surrounding text.
- <sub>: Lowers text placement below the baseline to create a **subscript** (e.g., H_2O).
- <sup>: Raises text placement above the baseline to create a **superscript** (e.g., $E = mc^2$).

6. Highlighting and Citations

- <mark>: Highlights text with a bright background color (typically yellow).

c. List Tags

HTML list tags are elements used to group a set of related items together in a structured format.

1. Unordered List (<ul>)

Use this format when the sequence or order of the items does not matter.

It displays items with bullet points by default.

- **<ul>**: Marks the beginning and end of the entire unordered list container.
- **<li>**: Represents a single list item inside the wrapper.

2. Ordered List (<ol>)

Use this format when items must follow a strict, sequential orde. It numbers items automatically.

- **<ol>**: Marks the beginning and end of the entire sequential list container.
- **<li>**: Holds the individual entries within the wrapper.

3. Definition List

In this type of listing, definition can be listed.

- <dl>: The wrapper element that defines

the start and end of the description list.

- <dt>: The description term element which contains the word or phrase being defined.
- <dd>: The description details element which holds the actual definition or value.

d. Image and Image Maps

An **HTML Image Map** allows you to create multiple clickable areas on a single image, with each area linking to a different destination.

1. The <img> Element (The Base Image)

The graphic serves as the visual background for your map. It requires a unique attribute to attach itself to the coordinate data.

- **src**: The directory path or URL of your image asset.
- **alt**: Descriptive alternative text for web accessibility and screen readers.

- **usemap**: Points directly to the <map> element. Its value **must start with a hash symbol (#)**, followed by the identical value written in the map's name attribute.

2. The <map> Element (The Blueprint Container)

This tag acts as a wrapper that holds all your custom hotspot parameters together.

- **name**: Defines the map ID so the image tag can locate it. The value must perfectly match the usemap text (excluding the hash). Spaces are not allowed.

3. The <area> Element (The Hotspot Definition)

Placed entirely inside the <map> tags, each <area> creates a standalone clickable path.

- **href**: The target URL or path to follow when someone clicks inside this zone.

- **alt**: Alternative text explaining the individual link destination for screen readers.
- **shape**: Outlines the geometric profile of the hotspot. Accepts three configurations:
  rect (Rectangle)
  circle (Circle)
  poly (Polygon/Multi-sided shape)
- **coords**: Coordinates in pixels relative to the top-left edge of the image (0,0). The syntax varies by shape.

PRACTICAL 2

Design a webpage that makes use of

a. Table Tags

In HTML, the **<table>** tag serves as the structural container used to organize and present tabular data in a grid-like format of rows and columns.

- **<table>**: Defines the boundary of the entire table.
- **<tr>**: Stands for *Table Row*; it groups cell contents into a horizontal row.
- **<th>**: Stands for *Table Header*; defines header cells which are **bold** and **centered** by default.
- **<td>**: Stands for *Table Data*; defines regular grid cells that hold text, links, or images.

b. Form Tags

The **<form> tag in HTML** is used to create an interactive section on a webpage that collects user input and sends it to a web

server for processing.

- <label>: Provides a clickable text description linked to an input field for better accessibility.
- <input>: Creates interactive fields like text boxes, checkboxes, radio buttons, or passwords depending on its type attribute.
- <textarea>: Generates a multi-line box for longer text inputs like comments.
- <select>: Creates a dropdown list of options.
- <button type="submit">: Submits the completed form data to the server.

c. Navigation across multiple pages

Navigation across multiple pages in HTML refers to the mechanism of connecting different web documents (HTML files) together using hyperlinks, allowing users to move seamlessly from one page to another within a website.

- **<nav> Element**: Defines a block of major navigation links for accessibility and structure.
- **<a> (Anchor) Tag**: The functional hyperlink element that connects the pages.
- **href Attribute**: The critical attribute inside the anchor tag specifying the file destination path.

d. Embedded Multimedia Element

Embedded multimedia elements in HTML are specialized tags used to integrate external media—such as audio, video, vector graphics, and interactive plug-ins—directly into a web page. Rather than just linking to a file, these elements render and play the media directly within the browser layout.

1. Core HTML Multimedia Elements

- **<video>**: Integrates playable video files directly on the page. Supported native

formats include MP4, WebM, and Ogg.

- **<audio>**: Embeds sound content like music, podcasts, or audio clips. Supported native formats include MP3, WAV, and Ogg.
- **<img>**: Displays static or animated images (JPEG, PNG, GIF, SVG).
- **<iframe>**: Embeds an entire external web page or third-party players like YouTube videos.

2. Key Multimedia Attributes

To control how these elements behave, HTML utilizes specific properties within the tags:

- **src**: Defines the path or URL of the media file.
- **controls**: Displays native playback buttons like play, pause, and volume control.
- **autoplay**: Starts playing the media instantly when the page finishes

loading.

- **loop**: Automatically restarts the media from the beginning once it ends.
- **width / height**: Sets the exact display dimensions of the media player container.

PRACTICAL 3

Create an HTML page using tags to accomplish the following:

a. A paragraph containing text "All that glitters is not gold". Bold face and italicize this text

Paragraph Tag

The **HTML <p> tag** defines a paragraph of text. It is a block-level element, which means it **always starts on a new line** and browsers automatically add a small vertical margin before and after it.

Strong and Emphasis tags are used

b. Create equation:

$x=1/3(y1^2+z1^2)$

To create this equation, <p> and <sup> are used

c. Put a background image to a page and demonstrate all attributes of background image

Inline CSS

Inline css is used to apply unique styles directly to a single HTML element by using the style attribute inside the opening HTML tag.

To apply inline CSS, add style="property: value;" inside the HTML tag. Separate multiple properties with a semicolon (;).

Background attribute is placed in the <body> tag because the body represents the visible content area of the webpage. Placing it there ensured the image or color covered the entire viewport.

d. Create unordered list of 5 fruits and ordered list of 3 flowers

Unordered list tag and ordered list tag are used to make list.

PRACTICAL 4

4. Create the following table using HTML tags. Properly align cells, give suitable cell padding and cell spacing, and apply background color, bold and emphasis necessary

Table Tag is used to form the table in which <tr>, <td> and <th> tags used its attributes.

- To change an HTML table's background color use the **bgcolor attribute** directly inside the table tags.
- **cellpadding**: This attribute is used in table tag only. Controls the internal space between the cell content and the cell border.
- **cellspacing**: This attribute is used in table tag only. Controls the external space between adjacent table cells.
- colspan: Dictates how many columns a single cell should stretch across

horizontally.

- rowspan: Dictates how many rows a single cell should stretch down vertically.
- align: This attribute position the text to center, left or right.

Bold<b> and Emphasis Tags are also used

PRACTICAL 5

5. Use HTML5 for performing following tasks:

a. Draw a square using HTML5 SVG, fill the square with green color and make 6px brown stroke width

The HTML **<svg> tag** acts as a container for **Scalable Vector Graphics**, allowing you to embed 2D vector shapes, paths, text, and animations directly into a web page.

- **width and height**: Sets the display dimensions of the SVG container on the webpage.
- **viewBox**: Defines the internal aspect ratio and coordinate system grid
- **xmlns**: Specifies the XML namespace (http://www.w3.org/2000/svg). It is mandatory for standalone .svg files but optional when embedding inline within HTML5 documents.
- **<rect>**: Draws rectangles and squares

using x, y, width, and height properties.

- **fill:** Specifies the solid color, gradient reference, or pattern applied to the inside area of a shape.
- **stroke:** Outlines the perimeter edge with a specified color target.
- **stroke-width:** Sets the stroke thickness in absolute units or relative scaling.

b. Write the following mathematical expression by using HTMI 5 MathML.:

$d=x^2-y^2$

This code consists of <p> and <sup>

c. Redirecting current page to another page after 5 seconds using HTML5 meta tag

An HTML <meta> tag provides metadata (information about data) for a web page. It is placed strictly inside the <head> section of the document, remains invisible to website visitors, and is parsed by browsers, search engines, and web

crawlers.

- **Character Encoding**: Specifies the text encoding (almost always UTF-8) to render letters, emoji, and symbols correctly.
- **Page Refresh**: Automatically reloads or redirects the page after a set number of seconds

Paragraph<p> and Anchor<a> tag are used as an alternative method through link

Practical 6

c. Demonstrate the following HTML5 Semantic tags- <article>, <aside>, <details>, <figcaption>, <figure>, <footer>, <header>, <main>, <mark>, <section> for a webpage that gives information about travel experience

- **<details> tag:** The **<details> tag** in HTML creates a **clickable dropdown menu** that hides information until a user clicks to reveal it.
- **<figure> tag:** The **<figure> tag** in HTML is a container used to **group an image (or other media) and its caption together** as a single unit.
- **<figcaption> tag:** The **<figcaption> tag** is used to add a **visible text caption** to an image, diagram, or chart on a webpage.

Practical 7

Design a webpage that make use of Cascading Style sheets with

To connect external CSS to an HTML document, you must place a **<link> tag inside the <head> section** of your HTML file.

a. CSS properties to change the background of a Page.

- **background**: Acts as a shorthand property to combine all the above rules into one single line.
- **background-size**: Controls image sizing, using cover to span the entire screen or contain to fit inside it.
- **background-position**: Aligns the image using coordinates or keywords like center or top left.

b. CSS properties to change Fonts and Text Styles

- **font-family**: Sets the typeface (e.g.,

Arial, sans-serif).

- **font-weight**: Sets boldness using keywords (bold) or numbers .
- **color**: Modifies the text color using HEX, RGB, or named values.
- **text-align**: Aligns layout bounds via left, right, center, or justify.
- **text-decoration**: Adds lines like underline, line-through, or strips default styles (none).

CSS properties for positioning an element

- **top**: Moves the element relative to the upper edge.
- **bottom**: Moves the element relative to the lower edge.
- **left**: Moves the element relative to the left edge.
- **right**: Offsets the element from the right edge.

Practical 8

Using CSS

a. Create a class called income, and make it a background color of #0ff.

b. Create a class called expenses, and make it a background color of #f0f.

c. Create a class called profit, and make it a background color of #f00.

d. Throughout the document, any text that mentions income, expenses, or profit, attach the appropriate class to that piece of text.

Further create following line of text in the same document: The current price is 50 and new price is 40

Internal CSS (also called embedded CSS) is a method used to style a single HTML document by placing all CSS rules directly inside a <style> tag within the <head> section of the page.

In Cascading Style Sheets (CSS), a **class** is

a reusable attribute used to select and apply specific styling rules to one or multiple HTML elements at the same time.

- tag: The tag stands for **"division"** and acts as an **invisible box or container** used to group other elements together.
- tag is an empty container used to target and style a specific part of a text without breaking the layout or starting a new line.

Practical 9

Using CSS

a. Change the tag li to have the following properties:

□A display status of inline

□A medium, double-lined, black border

□No list style type

b. Add the following properties to the style for li:

□Margin of 5px

□Padding of 10px to the top, 20px to the right, 10px to the bottom, and 20px to the left

Practical 10

JavaScript in HTML is a scripting language used to add interactivity, logic, and dynamic behaviors to a static web page.

a. Using JavaScript design, a web page that prints factorial/Fibonacci series/any given series.

b. Design a form and validate all the controls placed on the form using Java Script.

Write a JavaScript program to display all the prime numbers between 1 and 100.

Practical 11

a. Write a JavaScript program to accept a number from the user and display the sum of its digits.

b. Write a program in JavaScript to accept a sentence from the user and display the number of words in it. (Do not use split () function).

c. Write a java script program to design simple calculator.

Practical 12

Using JavaScript

a. Design a web page demonstrating different conditional statements.

b. Design a web page demonstrating different looping statements.